International Cybersecurity and Digital Forensics Academy (ICDFA)



Course Code: CIP-B101

Course Name: Basic Computer Skills for Digital Forensics

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Lab Submission Report

Lab Number: Lab 1B

Lab Name: PC System Fundamentals, Disk Sectors and Partition Analysis

Date of Submission: 7TH November 2026

Platforms Used: Kali Linux (Purple) and Microsoft Windows 11 Pro

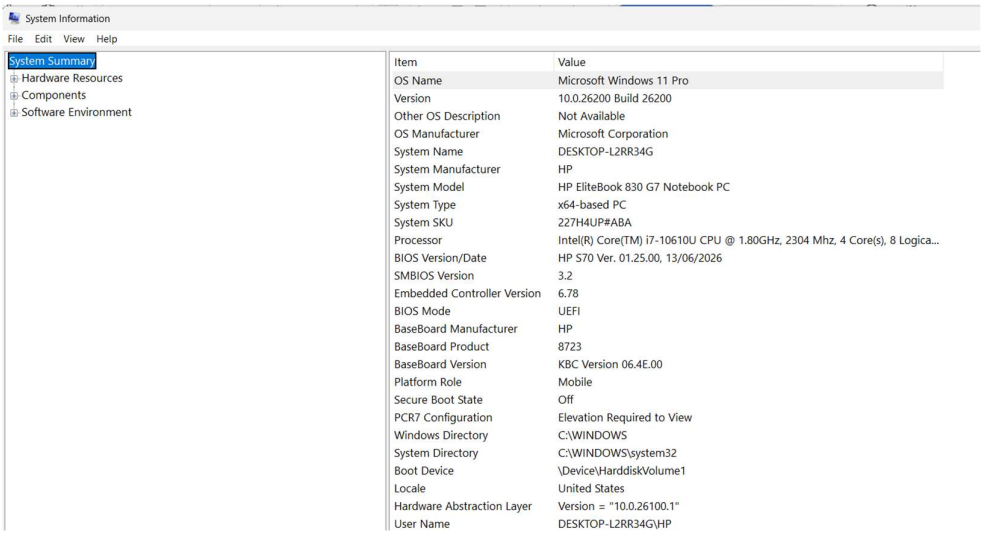
Tools Used: WinHEX

Part A - System Identification and Evidence Layer Mapping

In this section, you will identify the major computer system layers and explain how each layer can produce forensic evidence.

A1. Capture basic system specification

On Windows, open System Information by pressing Windows + R, typing msinfo32, and pressing Enter.



Also run the command below in Command Prompt and capture the first page of output.

```
C:\Windows\System32>systeminfo

Host Name:                 DESKTOP-L2RR34G
OS Name:                   Microsoft Windows 11 Pro
OS Version:                10.0.26200 N/A Build 26200
OS Manufacturer:           Microsoft Corporation
OS Configuration:          Standalone Workstation
OS Build Type:             Multiprocessor Free
Registered Owner:          HP
Registered Organization:
Product ID:                00330-53958-70151-AAOEM
Original Install Date:     18/06/2026, 14:15:20
System Boot Time:          06/11/2026, 08:56:56
System Manufacturer:       HP
System Model:              HP EliteBook 830 G7 Notebook PC
System Type:               x64-based PC
Processor(s):              1 Processor(s) Installed.
                           [01]: Intel64 Family 6 Model 142 Stepping 12 GenuineIntel ~1502 Mhz
BIOS Version:              HP S70 Ver. 01.25.00, 13/06/2026
Windows Directory:         C:\WINDOWS
System Directory:          C:\WINDOWS\system32
Boot Device:               \Device\HarddiskVolume1
System Locale:             en-us;English (United States)
Input Locale:              en-us;English (United States)
Time Zone:                 (UTC+03:00) Nairobi
Total Physical Memory:     16,157 MB
Available Physical Memory: 5,202 MB
Virtual Memory: Max Size:  21,277 MB
Virtual Memory: Available: 7,470 MB
Virtual Memory: In Use:    13,807 MB
Page File Location(s):     C:\pagefile.sys
```

A2. Capture storage information

Open Command Prompt or PowerShell.

```
PS C:\WINDOWS\system32> Get-Disk

Number Friendly Name Serial Number                    HealthStatus         OperationalStatus      Total Size Partition
                                                                                                             Style
------ ------------- -------------                    ------------         -----------------      ---------- ----------
0      PM9A1 NVMe... 3648_3330_52C3_3434_0025_3858... Healthy              Online                  476.94 GB GPT


PS C:\WINDOWS\system32> Get-Partition


   DiskPath: \\?\scsi#disk&ven_nvme&prod_pm9a1_nvme_samsu#5&37df2d22&0&000000#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}

PartitionNumber  DriveLetter Offset                                        Size Type
---------------  ----------- ------                                        ---- ----
1                            1048576                                    100 MB System
2                            105906176                                   16 MB Reserved
3                C           122683392                                 475.65 GB Basic
4                            510850498560                                1.17 GB Recovery


PS C:\WINDOWS\system32> Get-Volume

DriveLetter FriendlyName FileSystemType DriveType HealthStatus OperationalStatus SizeRemaining      Size
----------- ------------ -------------- --------- ------------ ----------------- -------------      ----
                         NTFS           Fixed     Healthy      OK                    84.63 MB   1.17 GB
C                        NTFS           Fixed     Healthy      OK                   312.81 GB 475.65 GB
```

A3. Map evidence sources to system layers

Create a table in your report showing at least five system layers/sources and the type of evidence that can be generated from each one. Use this guide: User activities = Applications + Operating System + Disk/Memory.

Layer / Source	Example Evidence	Where It May Be Found	Why It Matters
User layer	Login Sessions, User Account Activity such as number of followers, posts shared, accessed websites	Browser Profiles, User Account Activity Records	Links user action to system activity and can be correlated with other evidence e.g. timestamps
Application layer	Metadata from social media apps, app usage history	SQLite Databases, application logs	Shows application event activities that can be used to generate evidence
OS layer	Login Events, Scheduled Processes	Windows Registry, Sysmon Logs, Linux Event Logs	Shows system events and account activity
Disk layer	File Creation Timestamps, File system metadata, data streams	Filesystem metadata	Used to trace deleted or modified data
Memory/process layer	Command and PowerShell Processes, Loaded Dynamic Link Libraries (DLLs)	Process Listings, RAM Captures	Shows live system behavior and is used to identify running processes and whether they were suspicious

Explain why evidence may exist in more than one layer.
This is because activities done digitally affect different layers. For example, the OS Layer captures the Login Event for an individual but if the user uses a Remote Desktop Service, the application used also generates an associated log with the relevant timestamp. This is useful as it enables investigators to correlate timelines instead of depending on one source of evidence.

Part B - Program, Process and Thread Investigation

B1. Observe running processes

Capture a screenshot showing running processes.

```
PS C:\WINDOWS\system32> Get-Process | Select-Object -First 20 Name,Id,CPU,WorkingSet

Name                     Id CPU        WorkingSet
----                     -- ---        ----------
AggregatorHost         8140             12255232
ai                    11256 1.828125     22450176
ai                    21028 0.71875      40566784
AppActions            21252 1.234375     42131456
AppHelperCap           3248              27930624
ApplicationFrameHost  14580 0.828125     50647040
audiodg                1516 783.265625   26226688
backgroundTaskHost     8020 2.203125     60669952
backgroundTaskHost    24624 0.203125     18399232
brave                   828 215.421875  392040448
brave                  4904 690.578125  112762880
brave                 11800 165.015625   63889408
brave                 12632 0.1875        9523200
brave                 13924 4.578125    137404416
brave                 16604 349.296875  210706432
brave                 17244 3728.84375  558190592
brave                 17352 9.53125      19910656
brave                 17812 150.6875     23408640
brave                 17820 4.828125     43941888
brave                 19088 3.515625     83091456
```

Select three processes and record their process names, PID values, and likely purpose.

Process Name	PID Value	Purpose of the Process
brave	828	Brave Update Service; process running a single tab
AppHelperCap	3248	HP Help Ap FSA Service
audiodg	8020	It is the Windows Audio Device Graph Isolation Process: manages spatial sound and third party sound adjustments

B2. Associated Questions

In your report, explain the difference between a program, a process, and a thread in your own words.

A program is a process in execution while a process is an instance of a program to be executed. A thread is a sub-process, and multiple are created and executed independently and at the same time when a program is running.

Explain why process information may be useful during malware analysis or live forensic triage.

Useful information such as Process Names, Process IDs are essential in finding out what was running, if it was intended (may have been a trojan or a polymorphic virus), timeframe of execution. This is important when investigating an incident and performing initial triage and containment.

Part C - Disk Sector and Disk Size Calculation

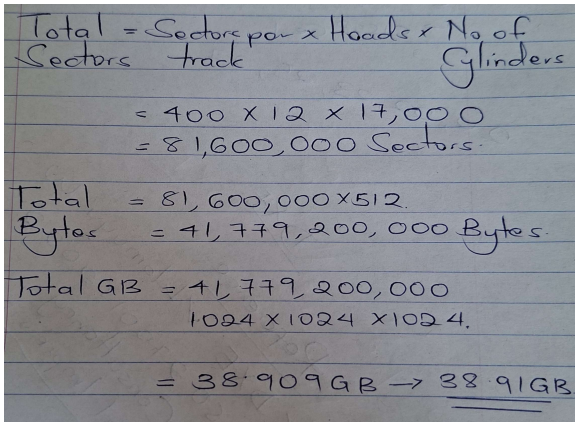
C1. Reproduce the slide calculation

Use the values from the slide to calculate the total disk size: sectors per track = 400, heads = 12, cylinders = 17000, and bytes per sector = 512.

Item	Formula / Value	Answer
Total sectors	400 x 12 x 17000	81,600,000 sectors
Total bytes	81,600,000 x 512	41,779,200,000 bytes
Kilobytes	41,779,200,000 / 1024	40,800,000 KB
Megabytes	40,800,000 / 1024	39,843.75 MB

Item	Formula / Value	Answer
Gigabytes	39,843.75 / 1024	38.91 GB

Actual Calculation Evidence



C2. Calculate your own disk or partition size

Use your partition table or disk details to identify at least two partitions.

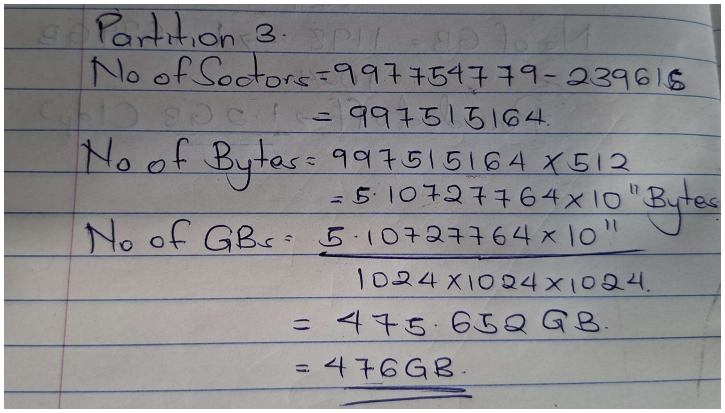
Partitioning style: GPT

Name	Ext.	Size	Created	Modified	Record changed	Attr.	1st sector
Start sectors		1.0 MB					0
Partition 1	FAT32	100 MB				H	2,048
Partition 2		16.0 MB				H	206,848
Partition 3 (C:)	NTFS	476 GB					239,616
Partition 4	NTFS	1.2 GB				SH	997,754,880
Unpartitioned space		3.3 MB					1,000,208,384

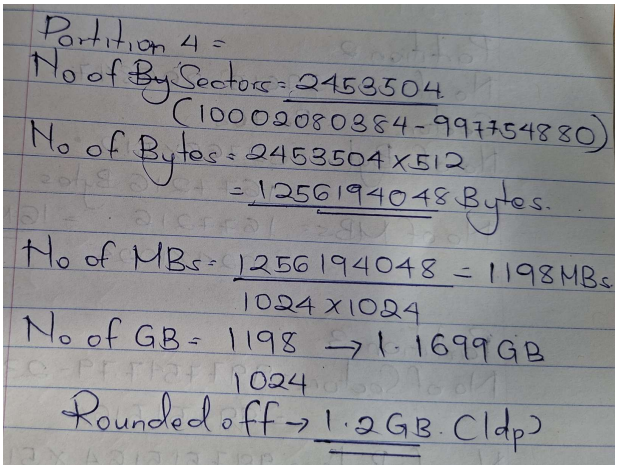
For each partition, record the start sector, number of sectors, sector size, and file system where available.

Partition Name	Start Sector	End Sector	Number of Sectors	Size(GB)	File System
Partition 3	239616	997754779	997515164	476GB	NTFS
Partiton 4	997754880	1000208384	1256194048	1.2GB	NTFS

Calculate the size of each partition in GB using the formula above.



Manual Calculation of Partition 3



Manual Calculation for Partition 4

Part D - Safe Partition Table Inspection

Linux/Kali method - read-only inspection

- Open Terminal in your Linux/Kali VM or host.
- Run the commands below and capture screenshots showing disk, partition, size, and file system information.
- Do not run mkfs, fdisk write commands, parted write commands, dd write commands, or mount read-write commands for this lab.

Supporting Screenshots

```
┌──(wamuyu㉿wamuyu)-[~/cip-b101-lab1a]
└─$ lsblk -o NAME,SIZE,TYPE,FSTYPE,MOUNTPOINT
NAME    SIZE TYPE FSTYPE MOUNTPOINT
sda    69.4G disk
├─sda1 65.8G part ext4   /
├─sda2    1K part
└─sda5  3.6G part swap   [SWAP]
sr0    1024M rom

┌──(wamuyu㉿wamuyu)-[~/cip-b101-lab1a]
└─$ sudo fdisk -l
[sudo] password for wamuyu:
Disk /dev/sda: 69.41 GiB, 74531536896 bytes, 145569408 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xb6f7e417

Device     Boot     Start       End   Sectors  Size Id Type
/dev/sda1  *         2048 138078207 138076160 65.8G 83 Linux
/dev/sda2       138080254 145567743   7487490  3.6G  f W95 Ext'd (LBA)
/dev/sda5       138080256 145567743   7487488  3.6G 82 Linux swap / Solaris
```

```
┌──(wamuyu㉿wamuyu)-[~/cip-b101-lab1a]
└─$ sudo parted -l
Model: ATA VBOX HARDDISK (scsi)
Disk /dev/sda: 74.5GB
Sector size (logical/physical): 512B/512B
Partition Table: msdos
Disk Flags:

Number  Start   End     Size    Type      File system     Flags
 1      1049kB  70.7GB  70.7GB  primary   ext4            boot
 2      70.7GB  74.5GB  3834MB  extended                  lba
 5      70.7GB  74.5GB  3834MB  logical   linux-swap(v1)  swap


┌──(wamuyu㉿wamuyu)-[~/cip-b101-lab1a]
└─$ blkid
```

```
┌──(wamuyu㉿wamuyu)-[~/cip-b101-lab1a]
└─$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            2.9G     0  2.9G   0% /dev
tmpfs           590M  1.1M  589M   1% /run
/dev/sda1        65G   14G   48G  23% /
tmpfs           2.9G  4.0K  2.9G   1% /dev/shm
none            1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
tmpfs           2.9G   36K  2.9G   1% /tmp
none            1.0M     0  1.0M   0% /run/credentials/getty@tty1.service
tmpfs           590M  100K  590M   1% /run/user/1000
```

Part E - MBR, GPT and Boot Process Explanation

E1. Explain the boot process

1. **Create a numbered list explaining how a PC boots.**
    a. The computer is Powered On
    b. The Power on Self-Test (POST) runs and the boot sequence is governed by the BIOS/UEFI, where the boot parameters are stored in the CMOS.
    c. Once completed, the MBR takes over
    d. The MBR points to the Boot Record of the OS
    e. The boot records are two: the System Volume that preloads the OS through the Boot Manager and the Boot Volume that loads the OS through the OS Kernel
    f. The Kernel loads and the chosen Operating System takes over.
2. **State whether your own system uses BIOS/Legacy or UEFI based on system information evidence.**
    The System uses UEFI.

Proof of Evidence:

Item	Value
OS Name	Microsoft Windows 11 Pro
Version	10.0.26200 Build 26200
Other OS Description	Not Available
OS Manufacturer	Microsoft Corporation
System Name	DESKTOP-L2RR34G
System Manufacturer	HP
System Model	HP EliteBook 830 G7 Notebook PC
System Type	x64-based PC
System SKU	227H4UP#ABA
Processor	Intel(R) Core(TM) i7-10610U CPU @ 1.80GHz, 2304 Mhz, 4 Core(s), 8 Logica...
BIOS Version/Date	HP S70 Ver. 01.25.00, 13/06/2026
SMBIOS Version	3.2
Embedded Controller Version	6.78
BIOS Mode	UEFI
BaseBoard Manufacturer	HP
BaseBoard Product	8723
BaseBoard Version	KBC Version 06.4E.00

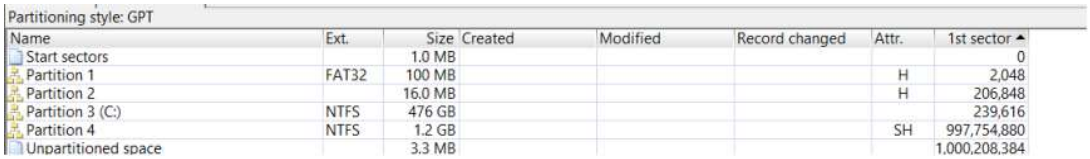
E2. Questions - Compare MBR and GPT

Answer

MBR is an older partition system that was common with earlier computer versions using BIOS/Legacy. It only supports 4 partitions and can only accommodate a total disk space that is less than 2 Terabytes. However, GPT is a more modern system that is associated with UEFI Systems, it supports large disk spaces of approximately 9728 TB. It has backup GPT headers which are useful in recovery and validation.

Part F - File System Identification and Interpretation

Identify the file system used by at least two volumes or partitions on your system.



Partitioning style: GPT

Name	Ext.	Size	Created	Modified	Record changed	Attr.	1st sector
Start sectors		1.0 MB					0
Partition 1	FAT32	100 MB				H	2,048
Partition 2		16.0 MB				H	206,848
Partition 3 (C:)	NTFS	476 GB					239,616
Partition 4	NTFS	1.2 GB				SH	997,754,880
Unpartitioned space		3.3 MB					1,000,208,384

For Windows systems, identify whether volumes use NTFS, FAT/FAT32, exFAT, or another file system.

Partition 1 uses FAT32 while Partitions 3 and 4 use NTFS.

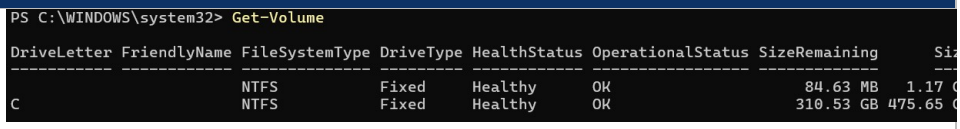
In your report, explain why a file system is important in digital forensics.

They determine how data is organized, stored and retrieved and hence, determines how investigation can be carried out. They are also a rich source of metadata as it holds information such as file name, size, timestamps, directory changes as well as manipulations and associated permissions that are extremely important in investigation. It also is useful in determine how data can be recovered for investigation and analysis. File Systems are also useful in influencing what type of recovery is done: APFS means that Mac OS extraction will take place, while NTFS will allow investigators to utilize Windows Forensics Tools

G - Final Practical Assignment

Practical Assignment Table

Task	Required Evidence	Student Notes / Answer
Identify partition 1	Screenshot of partition table and command output	PS C:\WINDOWS\system32> Get-Partition DiskPath: \\?\scsi#disk&ven_nvme&prod_pm9a1_nvme_samsu#5&37df2d22&0&000000#{53f56307-b6bf-11d0-94f2-00a0c91e PartitionNumber DriveLetter Offset Size Type ---------- ---------- ------ ---- ---- 1 1048576 100 MB System 2 105906176 16 MB Reserved 3 C 122683392 475.65 GB Basic 4 510850498560 1.17 GB Recovery
Record partition 1 sector count	Start sector, end sector or sector count	Start Sector: 239616 End Sector: 997754779 Sector Count = 997754880-239616=997515164 Sectors
Calculate partition 1 size	Formula and result in GB	No of Bytes= 997515164 * 512 = $5.10727764*10^{11}$ Bytes No of GB= $5.10727764*10^{1}$/(1024^3) =475.652GB
Identify partition 2	Screenshot of partition table and command output	PS C:\WINDOWS\system32> Get-Partition DiskPath: \\?\scsi#disk&ven_nvme&prod_pm9a1_nvme_samsu#5&37df2d22&0&000000#{53f56307-b6bf-11d0-94f2-00a0c91e PartitionNumber DriveLetter Offset Size Type ---------- ---------- ------ ---- ---- 1 1048576 100 MB System 2 105906176 16 MB Reserved 3 C 122683392 475.65 GB Basic 4 510850498560 1.17 GB Recovery
Record partition 2 sector count	Start sector, end sector or sector count	Start Sector: 997754880 End Sector: 10002080384 Sector Count = 10002080384-997754880=2453504 Sectors
Calculate partition 2 size	Formula and result in GB	No of Bytes= 2453504 * 512 = 1256194048 Bytes No of GB= 1256194048/(1024^3) =1.1699GB

Task	Required Evidence	Student Notes / Answer
Compare with OS-reported size	Disk Management, Get-Volume, lsblk, or df output	
Explain difference if any	Short explanation of rounding, reserved space, file system overhead, or GiB/GB differences	The difference between the OS-Reported Size and the calculated values is that the values are rounded off to 2 decimal places

Conclusion

Summarize what you learned and why this matters in digital forensics.

This lab has strengthened my fundamentals in how computers run, their specifications, how disk drives are structured, the boot process, how to inspect partitions as well as basic calculations to determine disk space. It also expanded my knowledge on various file systems and the various operating systems they are found in. These concepts matter because they form the foundation for Computer Forensics, understanding the concepts above are necessary if an investigator successfully wants to gather, analyze and correlate evidence at either the Disk Level or CPU Level. Understanding File Systems will enable an investigator to collect the appropriate evidence. Learning the different layers at which evidence is collected is important so that an investigator does not only rely on one source of evidence but can successfully correlate between all the 5 layers.